EDMeta: Exposure Distribution Meta-Regression Model

Statistical Model

Let $i = 1, \ldots, n$ index studies. Each study reports an estimated effect $\hat{\beta}_i$ with standard error $\hat{\sigma}_i$, along with a summary of its study-specific exposure distribution on $[\mathrm{l}_i, \mathrm{u}_i]$.

$$\hat{\beta}_i | \beta_i \overset{\text{ind}}{\sim} \text{Normal}\left(\beta_i,\ \hat{\sigma}_i^2\right), \quad i = 1, \ldots, n;$$

$$\beta_i | \mu, \eta(\cdot), \tau^2 \overset{\text{ind}}{\sim} \text{Normal}\left(\mu + \int_{\mathrm{l}_i}^{\mathrm{u}_i} \eta(x)\, g_i(x)\, \mathrm{d}x,\ \tau^2\right);$$

$$\int_{\mathrm{l}_i}^{\mathrm{u}_i} \eta(x)\, g_i(x)\, \mathrm{d}x \approx \mathbf{x}_i^{\mathrm{T}} \left(\widetilde{\mathbf{\Omega}}(\rho)^{-1} \tilde{\boldsymbol{\eta}}\right);$$

$$\mathrm{x}_{ij} = \frac{\mathrm{u}_i - \mathrm{l}_i}{m} \sum_{k=1}^{m} K(\mathrm{d}_{ik}, \mathrm{d}_j^*;\, \rho)\, g_i(\mathrm{d}_{ik}), \quad j = 1, \ldots, m^*;$$

- β_i: latent true effect for study i;
- μ: pooled mean effect;
- $\eta(\cdot)$: latent exposure-response function of primary interest;
- $\delta(x) = \mu + \eta(x)$: the exposure-response function reported in posterior inference;
- τ^2: between-study variance;
- $g_i(\cdot)$: study-specific exposure density on $[\mathrm{l}_i, \mathrm{u}_i]$; either Uniform$(\mathrm{l}_i, \mathrm{u}_i)$ (`exposure_distribution = 0`) or Truncated Normal with mean $\texttt{mid_exposure}_i$ and standard deviation $\texttt{sd_exposure}_i$, truncated to $[\mathrm{l}_i, \mathrm{u}_i]$ (`exposure_distribution = 1`);
- $\mathrm{d}_{i1}, \ldots, \mathrm{d}_{im}$: equally-spaced midpoint quadrature points across $[\mathrm{l}_i, \mathrm{u}_i]$, with $\mathrm{d}_{ik} = \mathrm{l}_i + \left(k - \frac{1}{2}\right) \frac{\mathrm{u}_i - \mathrm{l}_i}{m}$;
- $\mathrm{d}_1^*, \ldots, \mathrm{d}_{m^*}^*$: equally-spaced knots spanning $[\mathrm{l}^*, \mathrm{u}^*]$, where $\mathrm{l}^* = \min_i \mathrm{l}_i$ and $\mathrm{u}^* = \max_i \mathrm{u}_i$;
- $K(x, x';\, \rho) = \exp\left\{-\rho \frac{|x - x'|}{\mathrm{u}^*}\right\}$: exponential correlation function; $\rho > 0$ controls the rate of decay with exposure distance;
- $\mathbf{x}_i = (\mathrm{x}_{i1}, \ldots, \mathrm{x}_{im^*})^{\mathrm{T}}$: predictive process basis vector for study i;
- $\tilde{\boldsymbol{\eta}} = (\tilde{\eta}(\mathrm{d}_1^*), \ldots, \tilde{\eta}(\mathrm{d}_{m^*}^*))^{\mathrm{T}}$: GP values at the knot locations;
- $[\widetilde{\mathbf{\Omega}}(\rho)]_{jj'} = K(\mathrm{d}_j^*, \mathrm{d}_{j'}^*;\, \rho)$: $m^* \times m^*$ correlation matrix at the knots.

Prior Information

$\mu \sim \text{Normal}(0,\ 100^2)$;

$\tau^2 \sim \text{Inverse Gamma}(a_{\tau^2},\ b_{\tau^2})$;

- Default setting: $a_{\tau^2} = 0.01$, $b_{\tau^2} = 0.01$.

$\tilde{\boldsymbol{\eta}} | \psi^2 \sim \text{Normal}\left(\mathbf{0}_{m^*},\ \psi^2\, \widetilde{\mathbf{\Omega}}(\rho)\right)$;

$\psi^2 \sim \text{Inverse Gamma}(a_{\psi^2},\ b_{\psi^2})$;

- Default setting: $a_{\psi^2} = 0.01$, $b_{\psi^2} = 0.01$.

$\rho \sim \text{Gamma}(a_\rho,\ b_\rho)$;

- Default setting: $a_\rho = 1$, $b_\rho = 1$.

Default Initial Values

- $\beta_i = \hat{\beta}_i$ for all i;
- $\mu = 0$;
- $\tilde{\boldsymbol{\eta}} = \mathbf{0}_{m^*}$;
- $\tau^2 = 1.00$;
- $\psi^2 = 1.00$;
- $\rho = 1.00$.

Metropolis Settings

- ρ is updated via a Metropolis step on the log scale. The tuning parameter `metrop_sd_phi_eta_trans` controls the proposal standard deviation; target acceptance rate 20–40%.

Approximation Settings

- `m_approx` (m): number of midpoint quadrature points per study used to approximate the integral over $g_i(\cdot)$. Default: 100.
- `m_star_approx` (m^*): number of predictive process knots spanning $[\mathrm{l}^*, \mathrm{u}^*]$. Default: 100.